

The microclimate model

Day 2: Introduction to microclimate model

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Organized by:



The problem of scale – space

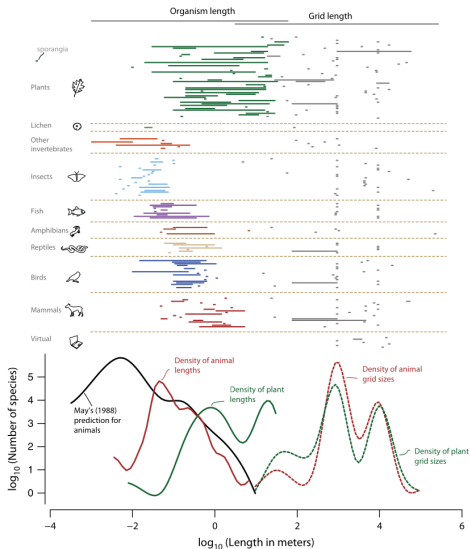


Figure 1: Species distribution models use environmental layers orders of magnitude greater than the organisms of interest (Potter et al., 2013)

The environment organisms actually experience

Weather stations measure conditions at ~1–2 m above ground.

Organisms live **at the surface** — on soil, under rocks, in leaf litter, buried in nests.

Definition

Microclimate is the physical environment (temperature, radiation, humidity, wind) at the organism's actual location.

Animals experience microclimates



Figure 2: *Salamandra salamandra* and *Alytes obstetricans* microclimates. Photos: Alfredo G. Nicieza

Vertical scale and organism experience

Different organisms live at different heights and depths — they experience different microclimates simultaneously:

- **Above ground (0–2 m):** Solar radiation, wind, shade dominate. *A basking lizard on a rock.*
- **Surface:** The critical interface where energy balance is computed. *A salamander under leaf litter.*
- **Shallow soil (0–10 cm):** Nests, eggs. Daily cycles, damped and lagged. *Reptile nest at 5 cm depth.*
- **Deep soil (10–200 cm):** Hibernation, burrows. Seasonal cycles with weeks–months lag. *Amphibian overwintering at 50 cm.*

Key insight

Two species at the same location can experience completely different microclimates if one is on the surface and the other is 10 cm below.

Climate is not enough

- Standard weather data captures conditions at ~ 2 m in an open field
- Organisms experience conditions near the surface, under vegetation, in burrows
- Microclimates determine:
 - Heat exchange and body temperature
 - Water balance and hydration
 - Behaviour (basking, sheltering, foraging)
 - Survival and reproduction

Core motivation

To predict how organisms respond to climate, we need microclimate — not just macroclimate.

The problem of scale – time

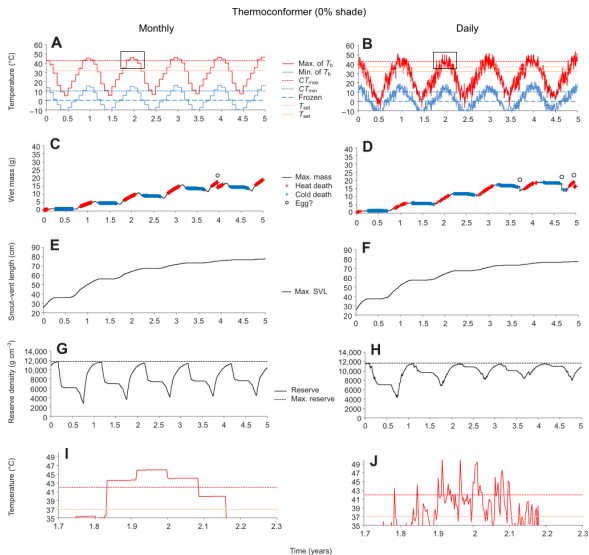


Figure 3: In addition to spatial scale, the temporal scale of predictor/forcing variables are also critical (Kearney et al., 2012)

The problem of scale – time

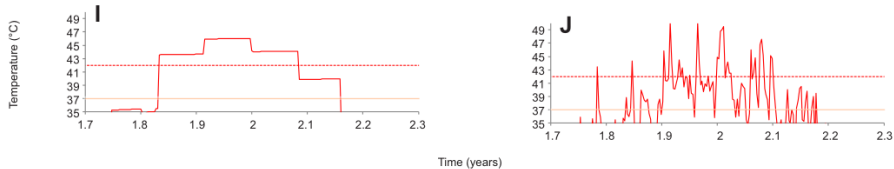


Figure 3: In addition to spatial scale, the temporal scale of predictor/forcing variables are also critical (Kearney et al., 2012)

What is NicheMapR?

- Developed by Kearney and Porter (2017).
- R package with a Fortran computational engine
- NicheMapR's microclimate model predicts hourly above- and below-ground conditions anywhere on Earth
- Inputs: weather, terrain, vegetation, soil properties
- Outputs: solar radiation, soil and air temperature, humidity, snow, soil moisture

Main strength

Mechanistic — predictions are physically based, so they remain valid under novel environmental conditions.

Two layers: R and Fortran

Fortran engine

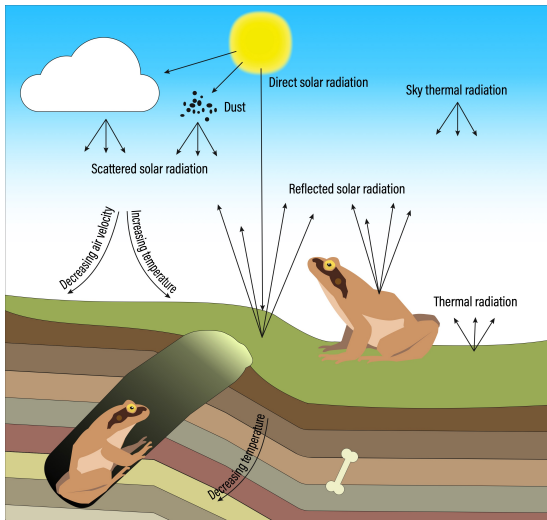
- Numerical computations
- Radiation physics
- Soil heat transfer
- Moisture dynamics
- Snow modelling

R interface

- Data preparation
- Global climate databases
- Running simulations
- Visualization
- User customization

The complexity of coupled ODEs requires Fortran's speed; R provides the flexible, user-friendly interface.

Biophysical Ecology processes



Solar radiation: the energy engine

Solar radiation is the primary energy input to the surface. NicheMapR computes how much energy reaches the ground at each hour by modelling its path through space and atmosphere.

The wavelength-specific irradiance on a flat surface is:

$$I_{\lambda} = S_{\lambda} \left(\frac{a}{r} \right)^2 \cos Z$$

where S_{λ} is the solar spectral irradiance, $(a/r)^2$ corrects for Earth-Sun distance, and Z is the zenith angle.

The zenith angle is derived from latitude, solar declination, and time of day. NicheMapR calculates radiation at 111 wavelengths and accounts for attenuation by Rayleigh scattering, aerosols, ozone, and water vapour.

Solar radiation: relevance for organisms

What this means on the ground

The solar radiation computation determines the daily and seasonal rhythm of energy input — when basking is possible, how intense it is, and how it varies with latitude and season.

Organism relevance:

- Basking ectotherms depend on solar radiation reaching the surface — cloud cover, shading, and slope aspect all modulate this
- Seasonal changes in solar angle determine the activity window for reptiles at different latitudes
- The spectral resolution (111 wavelengths) matters: UV and visible light penetrate differently through vegetation and water

Slope, aspect, and the microclimate mosaic

Most surfaces are not flat. A sloping surface receives different solar energy than a flat one:

$$Z_{SL} = \arccos \left(\cos(Z) \cos(SL) + \sin(Z) \sin(SL) \cos(AZ_{SUN} - AZ_{SL}) \right)$$

where SL is the slope angle and AZ_{SL} is the slope azimuth.

Organism relevance:

- A south-facing slope (northern hemisphere) receives more radiation — basking sites warm faster in spring
- A north-facing slope stays cooler and retains moisture — refuge for amphibians on hot days
- Adjacent slopes metres apart can differ in temperature by several degrees
- This drives fine-scale habitat selection and nest-site choice in reptiles

Atmospheric attenuation and elevation effects

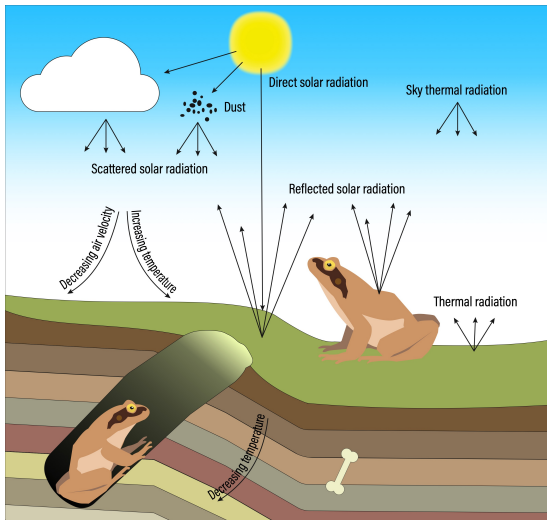
Solar radiation is attenuated by Rayleigh scattering, aerosols, ozone, and water vapour. At higher elevation, the thinner atmosphere scatters and absorbs less — so more radiation reaches the surface despite colder air.

Temperature decreases with elevation ($\sim 0.77^{\circ}\text{C}$ per 100 m for maxima, $\sim 0.39^{\circ}\text{C}$ for minima). The **freezing line** sets upper limits for freeze-intolerant species.

Organism relevance

A 1400 m elevation gain can reduce temperatures by $5\text{--}10^{\circ}\text{C}$ while increasing solar radiation — high-elevation organisms face colder air but more intense sunlight.

Biophysical Ecology processes



Longwave radiation: the invisible driver

All objects emit infrared radiation proportional to T^4 (Stefan-Boltzmann law). At night, longwave radiation dominates the energy budget.

Clear-sky emissivity depends on air temperature and vapour pressure:

$$\epsilon_{SKY} = 1.72 \left(\frac{e_A}{T_A} \right)^{\frac{1}{7}}$$

NicheMapR combines five longwave sources:

- **Clear-sky** emission from the atmosphere
- **Cloud** emission (clouds radiate as near-blackbodies)
- **Vegetation** radiation from surrounding canopy
- **Hill** radiation from surrounding terrain
- **Ground** emission from the surface itself

Longwave radiation and organism experience

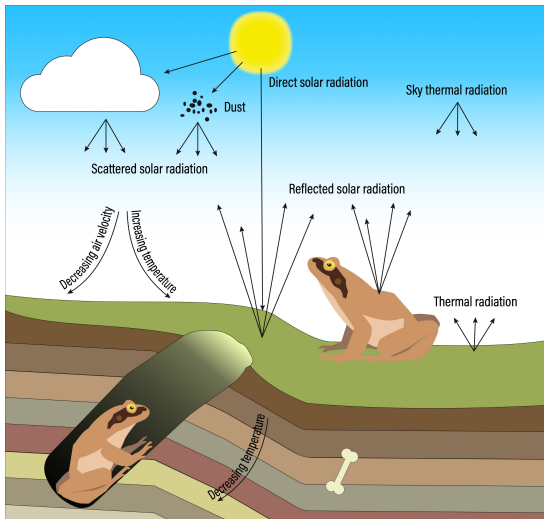
Nighttime cooling:

- Clear skies: the ground radiates to cold space — rapid cooling, frost risk
- Clouds: emit longwave back downward — slower cooling, warmer nights
- Canopy: blocks the view to cold sky — warmer nighttime microclimates

Organism relevance:

- Nocturnal amphibians depend on warm, humid nights for activity
- Nest placement under canopy vs. open sky changes nighttime minima by 5–10 °C
- Burrow walls radiate back at the occupant, buffering longwave losses
- Frost at ground level can kill eggs even when air temperature is above freezing

Biophysical Ecology processes



Soil types and albedo

NicheMapR includes 12 soil types. Three common examples:

- **Rock (type 0)**: high thermal conductivity — heats slowly, stores heat, small daily range
- **Sand (type 1)**: low conductivity and heat capacity — heats and cools rapidly, large daily range
- **Loam (type 4)**: intermediate properties

Soil albedo (REFL): Dark soil (REFL=0.05) absorbs most solar radiation and heats substantially; light soil (REFL=0.35) reflects more and stays cooler.

Practical relevance

Soil type and colour determine the thermal environment at nest depth — sand heats faster than loam, dark soil reaches higher peak temperatures.

Soil moisture and evaporative cooling

NicheMapR can simulate depth-specific soil moisture using Campbell's water balance model. It tracks infiltration, evaporation, transpiration, and redistribution through 19 moisture nodes, with hydraulic properties from the `CampNormTb19_1` table.

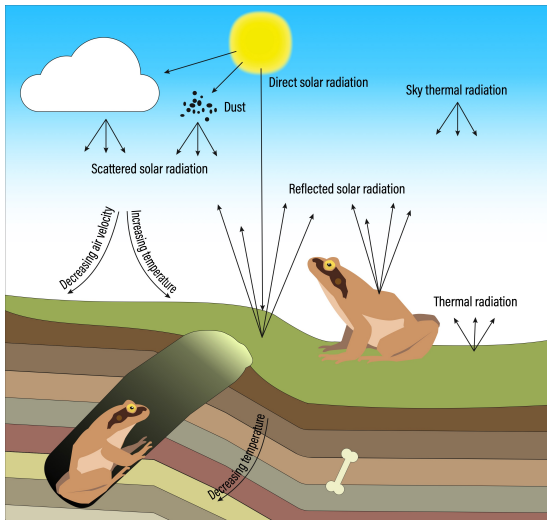
How moisture affects microclimate:

- Wet soil has higher heat capacity and thermal conductivity — dampens temperature extremes
- Evaporation consumes heat — wet surfaces stay cooler during the day
- Dry soil heats faster and reaches higher maximum temperatures

Organism relevance:

- Soil moisture determines habitat suitability for amphibians
- Water potential at nest depth affects egg hydration and survival
- Sand dries quickly near the surface but retains moisture at depth; loam retains moisture more evenly

Biophysical Ecology processes



From weather station to ground level

Standard weather data is measured at ~2 m. Organisms live near the surface. NicheMapR bridges this gap using boundary layer physics.

Wind speed decreases near the surface.

Air temperature increases towards the surface.

Why the boundary layer matters for organisms

The difference between 2 m and ground level is not just about height — it shapes how organisms experience their environment:

- A lizard at ground level may experience air 5–10 °C hotter than the 2 m reading — critical for activity and thermoregulation
- A salamander in leaf litter is in a nearly still air layer — minimal convection, high humidity, stable conditions
- Small ectotherms are tightly coupled to the microclimate at their specific height — moving 5 cm up or down can change body temperature substantially

Key scale insight

The difference between 1 mm and 1 m above ground can be as large as the difference between two climate zones.

The surface energy balance: everything comes together

The model finds a surface temperature T_{surf} that satisfies the balance of all energy fluxes:

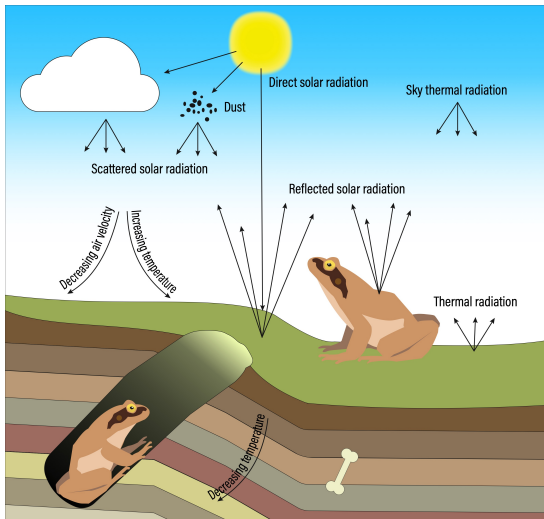
$$\dot{Q}_{solar} + \dot{Q}_{IR,in} - \dot{Q}_{IR,out} - \dot{Q}_{conv} - \dot{Q}_{evap} - \dot{Q}_{cond} = 0$$

Each term corresponds to a physical process organisms experience:

- $\dot{Q}_{solar}$: absorbed sunlight — basking, colouration, shade-seeking
- $\dot{Q}_{IR}$: net longwave — nighttime cooling, canopy buffering
- $\dot{Q}_{conv}$: convection — wind removes or adds heat
- $\dot{Q}_{evap}$: evaporation — evaporative cooling, moisture balance
- $\dot{Q}_{cond}$: soil conduction — heat storage, buffering

The model iterates this balance hourly, using bisection or Newton-Raphson, converging when T_{surf} closes the budget. This is how microclimate is generated at the temporal scale that matters for organisms.

Biophysical Ecology processes



Soil heat: the temporal dimension

Below the surface, temperature at depth z and time t is governed by the 1D heat conduction equation:

$$\frac{\partial T}{\partial t} = \frac{k}{\rho c} \frac{\partial^2 T}{\partial z^2}$$

Numerical implementation:

- 10 soil nodes, spaced more closely near the surface
- Adams predictor-corrector solver
- Thermal properties updated hourly based on soil moisture
- Bottom boundary set to mean annual air temperature

Depth creates lag and damping

- **Surface:** Full diurnal cycle, maximum amplitude, rapid response — *surface-active organisms*
- **2–5 cm:** Daily cycle, slightly damped, lagged by hours — *typical reptile egg depth, embryos experience daily fluctuations*
- **10–15 cm:** Amplitude halved, lagged by 6–12 hours — *deeper nests, more stable temperatures*
- **50–100 cm:** Diurnal cycle disappears, seasonal cycle dominates, lagged by weeks — *hibernation sites*
- **200 cm:** Nearly constant at mean annual temperature — *deep burrows, fully buffered*

Climate change implication

A 1 °C rise in air temperature shifts soil temperature at all depths. For shallow nests, this directly affects embryo development rate and survival.

Snow: insulation and albedo at high elevation

NicheMapR includes a snow module that adds up to 8 snow nodes above the soil surface. Snow has two main effects:

- **Insulation:** snow acts as a thermal blanket — ground stays warmer in winter and cooler in summer
- **Albedo:** fresh snow reflects most incoming solar radiation, reducing daytime heating

Realistic snowpack requires sequential daily simulation (`timeinterval=365`) over multiple years; the first year is discarded as spin-up. Output includes hourly snow depth (`SNOWDEP`).

Organism relevance

Snow insulates hibernating amphibians and reptiles from lethal winter temperatures. Losing snow cover due to climate change exposes overwintering life stages to freezing.

The integrative picture

Take-home message

Each component operates at different spatial and temporal scales. Together, they generate the microclimate that determines whether and how organisms persist.

Every organism — whether a basking lizard, a buried turtle egg, or a hibernating amphibian — experiences a unique combination of these components at its specific location, height, depth, and time of day.

Inputs and outputs

Inputs:

- Air temperature, relative humidity, wind speed
- Cloud cover, rainfall
- Soil properties: conductivity, density, water-holding capacity
- Terrain: slope, aspect, horizon angles
- Vegetation: shade fraction, leaf area index

Outputs:

- Hourly soil temperature at multiple depths
- Air temperature, relative humidity and wind speed at user-specified heights
- Solar and longwave radiation components
- Sky temperature, soil moisture, snow depth

Global workflow

- 1 **micro_global** (R): Aggregates climate (New et al., 2002), aerosols (GADS), and soil moisture (NOAA) data for any location
- 2 **microrun** (R interface): Passes parameters to the Fortran DLL
- 3 **Fortran engine**: Performs hourly iterations, solving the energy balance until T_{surf} converges

Key output matrices:

- metout: surface and air conditions, hourly
- soil: temperatures at 10 depths, hourly
- soilmoist: water content and potential at 19 depths

Global applicability

Users can run microclimate simulations anywhere on Earth using built-in global datasets.

The egg microclimate

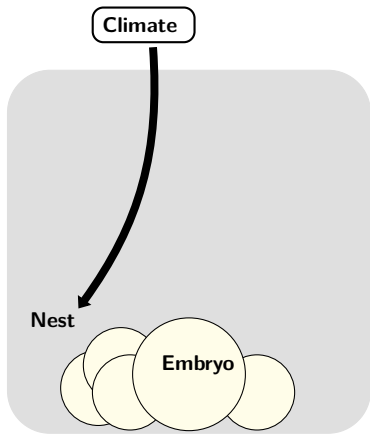


Figure 4: We can model the microclimates eggs experience at relevant temporal and spatial scales. Based on Mitchell et al. (2016)

The egg microclimate

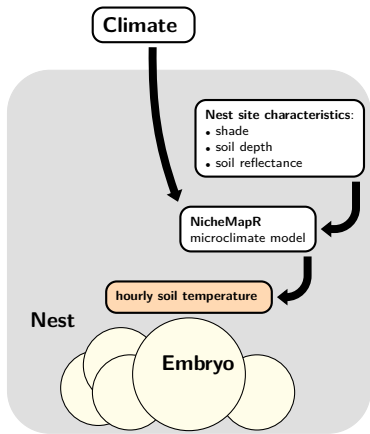


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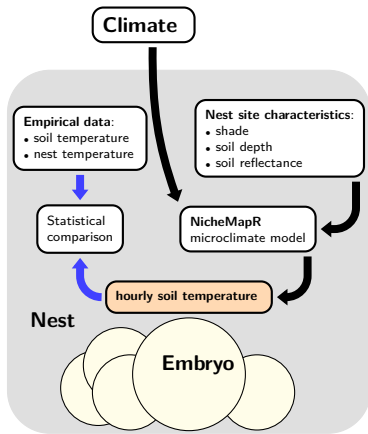


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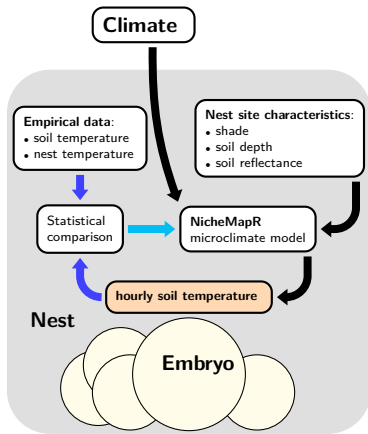


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Egg translocations informed by microclimate

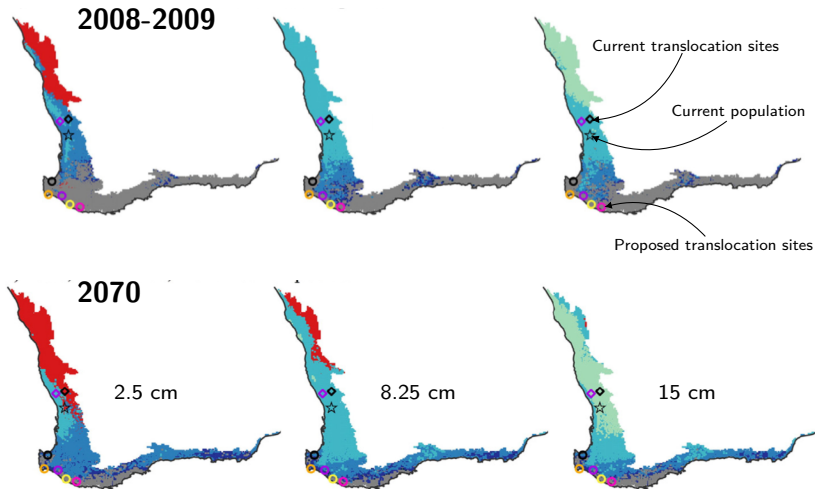


Figure 5: Simulated egg performance. Modified from Mitchell et al. (2016)

Calibrating microclimates



Example: egg
translocation

Calibrating microclimates

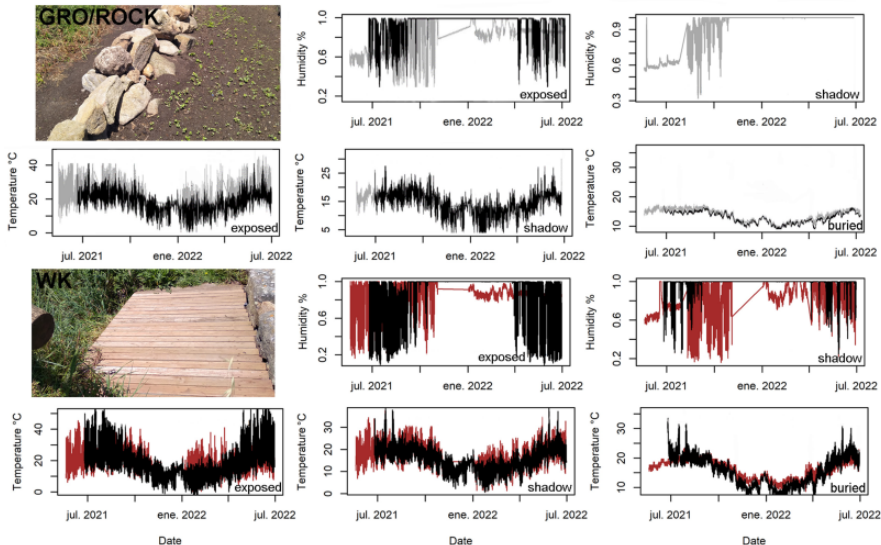


Figure 6: Calibrating microclimates, taken from Reyes-Puig et al. (2026)

Strengths and limitations

Strengths:

- **Mechanistic:** physically based — valid under novel climates
- **Fine resolution:** hourly output captures diurnal cycles critical for organisms
- **Integrated:** combines solar, longwave, convection, soil heat, and moisture in one framework
- **Proven:** extensively validated against field observations

Limitations:

- **Point-based:** no lateral heat flow — represents one vertical column
- **Data-hungry:** requires many parameters; sensitive to input quality
- **No vegetation dynamics:** vegetation structure is prescribed, not predicted

What NicheMapR enables

- Predicting microclimate at the spatial and temporal scales that actually matter to organisms
- Connecting macroclimate (weather data) to organism experience (temperature at the nest, the leaf litter, the burrow)
- Mechanistic species distribution modelling — asking not just where species occur, but why and whether they will persist

Core insight

Organisms experience microclimates, not climate averages. To understand their biology and predict their fate, we must model the environment at their scale.

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